

Hyperledger Sawtooth Architecture: Permissioning Design

Hyperledger Sawtooth includes the ability to control validator and transactor permissions. This article describes permissioning in Hyperledger Sawtooth by providing a detailed list of the requirements, capabilities and user stories that help to understand permissioning design.



Sawtooth divides permissioning into two general groups:

- Transactor key permissioning controls that can submit transactions and batches, based on signing keys, and...
- Validator key permissioning controls, with which nodes are allowed to establish connections to the validator network.

Each requirement has a short description, the network scenario that the requirement supports and associated user stories. User stories are short statements about a requirement or capability that an actor desires. These short descriptions illustrate the need

for specific permission requirements. The design of Sawtooth permissioning includes both local validator configuration and network-wide on-chain permissioning.

- Local configuration allows a validator to limit who is allowed to submit batches and transactions directly to that validator. This is useful, for example, if a company is running its own validator, and wishes to allow only members of that company to submit transactions.
- On-chain configuration allows a Sawtooth network to enforce consistent permissioning rules. For example, in validator key

permissioning, all the validators on the network will reject the same node from peering, ensuring that only known nodes can join the network.

Validator network scenarios

Sawtooth is designed to support public, consortium and private networks. The following network scenarios are used in the discussion on capability requirements to illustrate when each requirement could be useful.

- **Public network:** In a public network, all connections are allowed, and all transactors are allowed to sign batches and transactions.